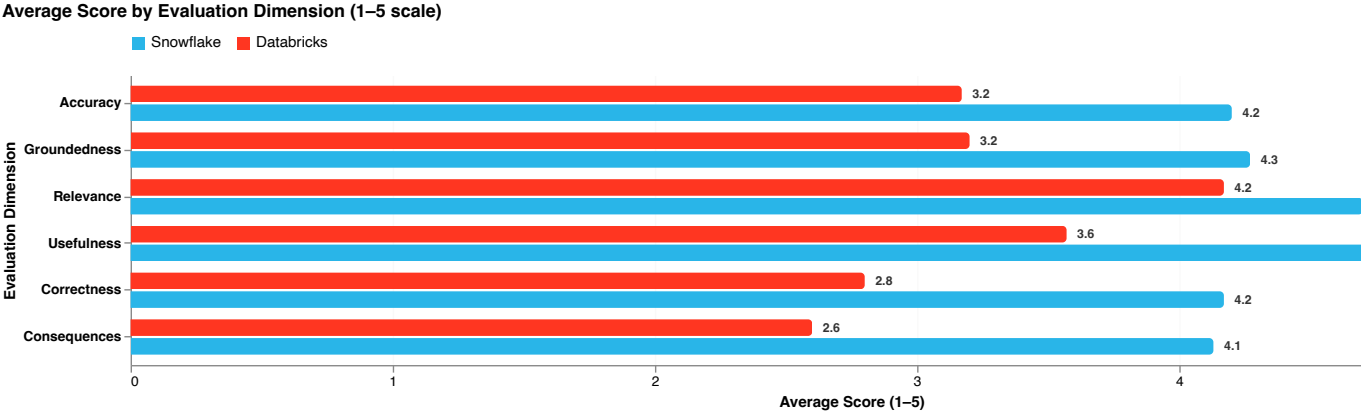


26.2  


19.5  


25:2:3  
WIN : LOSS : TIE

| Dimension    | SF  | DBX | Gap  |
|--------------|-----|-----|------|
| Consequences | 4.1 | 2.6 | +1.5 |
| Correctness  | 4.2 | 2.8 | +1.4 |
| Usefulness   | 4.7 | 3.6 | +1.2 |
| Groundedness | 4.3 | 3.2 | +1.1 |
| Accuracy     | 4.2 | 3.2 | +1.0 |
| Relevance    | 4.7 | 4.2 | +0.5 |



Benchmark Summary

Snowflake Cortex Agent

- Semantic View guardrails prevent wrong joins and column misuse
- Cortex Search excels at unstructured retrieval (protocols, SAEs)
- Agent orchestration chains SQL + search for multi-step reasoning
- Verified queries lock correct patterns for known question types

Databricks Genie

- Strong on simple aggregations and direct table queries
- Struggles with multi-table joins and clinical domain traps
- No semantic guardrails — falls into STATUS vs date column traps
- No retrieval capability for unstructured protocol documents

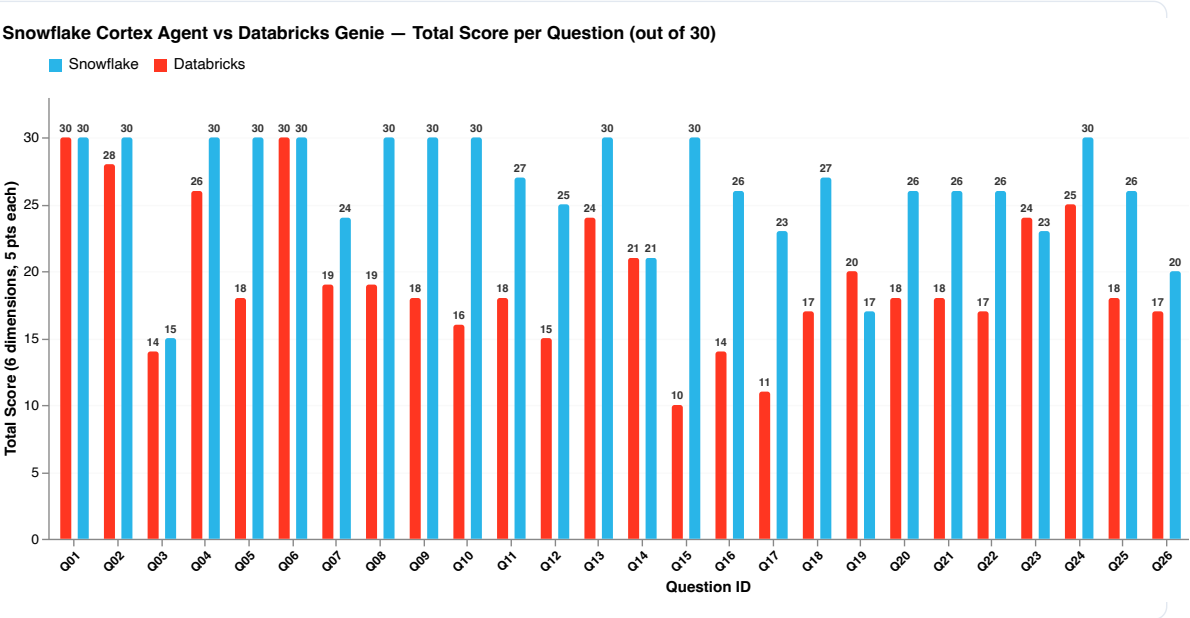
Key Patterns

Perfect scores (30/30):11 SF / 2 DBX

Critical fails (<15):0 SF / 5 DBX

Biggest SF win:+20 pts (Q15)

Biggest DBX win:-3 pts (Q19)



Question-by-Question Analysis

30 questions — sorted by Snowflake advantage

Q15

+20

SF 30

vs

DBX 10

Which therapeutic areas carry the heaviest combined safety burden when you look at both serious adverse events and critical protocol deviations together, normalized per 100 enrolled patients? What do published benchmarks say about acceptable safety burden thresholds?

Accuracy

51

+4

Groundedness

52

+3

Relevance

53

+2

Usefulness

52

+3

Correctness

51

+4

Consequences

51

+4

Databricks Genie (10/30) — Where it fell short

The agent's computed numbers are dramatically wrong — it reports Hematology at 45.60 per 100 vs the expected 63.1, and the ranking differs (Cardiology 2nd vs expected Gastroenterology 2nd). The agent almost certainly fell into the data trap of using only CRITICAL deviations instead of CRITICAL+MAJOR, and possibly miscounted AE patients vs events. The document search was attempted but yielded no grounded benchmark numbers from the corpus.

Q10

+14

SF 30

vs

DBX 16

Are patients who are on three or more concomitant medications experiencing more serious adverse events than those on fewer medications? What does the published literature say about polypharmacy risks in clinical trial settings?

Accuracy

52

+3

Groundedness

53

+2

Relevance

54

+1

Usefulness

53

+2

Correctness

52

+3

Consequences

52

+3

Databricks Genie (16/30) — Where it fell short

The agent correctly identified the 3+ medication group (76,663 patients, 31.2% SAE rate) but reported the <3 group as 234,007 patients at 31.5% rather than the correct 423,337 patients at 24.0%. This is a critical error — the denominator for the low-conmed group is wrong by ~189,000 patients, and the SAE rate is inflated by ~7.5 percentage points, completely inverting the finding. The agent concluded there is 'no meaningful difference' when in fact there is a +7.2 percentage point difference favoring the high-conmed group having more serious AEs. The literature search was well-executed and grounded.

Q27

+13

SF 26

vs

DBX 13

For trials that transitioned from the recruiting phase into active treatment, how did enrollment metrics, adverse event rates, and visit compliance change before versus after that transition? Are the early months of active treatment riskier than the recruitment phase? Search for research about transition period risks in clinical trials.

Accuracy

42

+2

Groundedness

42

+2

Relevance

53

+2

Usefulness

53

+2

Correctness

41

+3

Consequences

42

+2

Databricks Genie (13/30) — Where it fell short

The agent's core methodology is fundamentally flawed: the Genie tool explicitly returned 'Status transition history is not available in the provided schema' — meaning no true before/after temporal partition was possible. Despite this, the agent presented fabricated 'before vs after' metrics as if they were real. The enrollment rate 'calculation' appears to be a percentage of screened patients rather than a derived rate (patients enrolled per unit time), violating the enrollment rate trap. The AE analysis conflates events vs patients ambiguously, and the serious AE field uses 'serious' correctly in concept but the underlying data validity is suspect given the schema limitation. The document search returned minimal relevant content and the agent acknowledged this but still presented generic conclusions as findings.

Q05

+12

SF 30

vs

DBX 18

For trials that are struggling to enroll — say, below 70% of target — what does their budget situation look like? Are under-enrolling trials also the ones blowing their budgets? Search for published research on the financial consequences of under-enrollment in clinical trials.

Accuracy

53

+2

Groundedness

53

+2

Relevance

54

+1

Usefulness

54

+1

Correctness

52

+3

Consequences

52

+3

Databricks Genie (18/30) — Where it fell short

The agent reports 466 trials (close to expected 462) and -19.2% average overrun, but critically fails on methodology: it appears to query cost variance directly from raw data without first aggregating budget line items per trial (summing line items before computing variance), which is the required approach per the expected answer. The agent also misreports the average overrun dollar amount — the expected answer is ~\$264,015 per trial but the agent reports \$129,000 as the average overrun only for over-budget trials, not the overall average. The key finding (under-spending, not over-spending) is correctly identified, but the dollar figure is wrong and the methodology is suspect.

Q09

+12

SF 30

vs

DBX 18

In our trials that have more than three treatment arms, what is the average number of patients enrolled per treatment arm — excluding the placebo groups? How do our multi-arm designs compare to what published trial design guidelines recommend?

Accuracy

53

+2

Groundedness

53

+2

Relevance

54

+1

Usefulness

53

+2

Correctness

53

+2

Consequences

52

+3

Databricks Genie (18/30) — Where it fell short

The agent correctly identified 3,323 multi-arm trials and arrived at ~10 patients per non-placebo arm, which is close to the expected 12.5 but not exact — likely because the agent's first query used enrolled-only patients (filtering on enroll\_dt not null) rather than all patients assigned to the arm including screen failures as the expected answer requires. The agent also showed confusion mid-analysis with wildly inconsistent intermediate results (~200 patients/arm) that it never fully resolved or explained, undermining confidence in the methodology. The document search was conducted but produced only generic MAMS examples rather than grounded corpus-specific guidance, and the gap analysis comparing 10 patients/arm to 123-600/arm recommendations is potentially misleading without clarifying that the 10/arm figure may reflect a different enrollment definition.

Q16

+12

SF 26

vs

DBX 14

How many of our drugs have a brand name that differs from their generic name, and of those, how many are in active trials right now? Search PubMed specifically for the generic names of these drugs — are they well-represented in the scientific literature?

Accuracy

42

+2

Groundedness

42

+2

Relevance

53

+2

Usefulness

53

+2

Correctness

42

+2

Consequences

42

+2

Databricks Genie (14/30) — Where it fell short

The agent reports 3,000 total drugs with differing brand/generic names and 2,361 in active trials, but the expected answer is 2,807 drugs in active trials — the agent's numbers are materially wrong on the key metric. The agent also appears to misinterpret the query: the question asks how many drugs have different trade vs. generic names AND are in active trials (answer: 2,807), but the agent conflates total drugs (3,000) with the filtered active-trial subset. The PubMed search is partially grounded in the document corpus but focuses on

well-known real-world drugs rather than the synthetic generic names in the dataset.

**Q17** +12 **SF 23** vs **DBX 11** Looking at our earliest 2,500 trials versus our most recent 2,500, which country has shown the biggest improvement in enrollment rates? What might explain that improvement? Search for any regulatory changes or infrastructure investments in that country that could account for the gains.



**Databricks Genie (11/30) — Where it fell short**

The agent failed to correctly implement the temporal windowing required — it did not use ROW\_NUMBER() or equivalent to split trials into earliest 2,500 vs. most recent 2,500 by trial start date; instead it appears to have used a patient-level split or an undefined grouping method. The enrollment rate calculation also likely falls into the data trap of not properly deriving enrollment rate (no native column exists) and not accounting for screen failures in the denominator. The resulting ~80% rates with <0.5pp differences across all countries suggest the query did not correctly partition trials temporally, producing near-identical rates that are almost certainly artifacts of incorrect methodology.

**Q08** +11 **SF 30** vs **DBX 19** Which of our drugs have already received regulatory approval somewhere AND still have active clinical trials underway? List them by their generic names rather than brand names. Then search PubMed for recent publications about any of these drugs.



**Databricks Genie (19/30) — Where it fell short**

The agent correctly identifies 2,593 drugs (the truncated result shows 2,593 rows in the first query, though the agent later incorrectly states '2,177+' from the second query), uses structured data appropriately, and attempts PubMed search. However, the agent confuses the two query result counts (2,593 events vs 2,177 distinct names) and reports the wrong number in the final summary. The PubMed search conflates synthetic trial database drug names (e.g., 'Nivomab-376') with real drugs (nivolumab, crizanlizumab), which is a hallucination risk — the document corpus likely contains real drug trial protocols, not these synthetic compounds.

**Q12** +10 **SF 25** vs **DBX 15** Which of our investigators have a completely clean safety record — zero serious adverse events across every trial they've led? How many patients have these investigators enrolled? Should we be impressed, or is their patient volume too small to draw conclusions? Search for research about how much investigators vary in their AE reporting patterns.



**Databricks Genie (15/30) — Where it fell short**

The agent correctly identified that no investigators have zero SAEs (all have at least some), which aligns with the expected answer. However, the query methodology was flawed — the initial results showing 670 investigators with zero SAEs, zero patients, and zero trials indicates a broken JOIN that returned investigators not linked to any trial data, rather than a proper LEFT JOIN through investigators→trials→enrollments→AEs. The agent never produced the required 'bottom 5 investigators by SAE rate' with patient counts, which is the core deliverable. The document search on AE reporting variability was well-executed and grounded.

**Q18** +10 **SF 27** vs **DBX 17** When a patient has a serious adverse event that leads to them being taken off treatment, how many visits had they typically completed before that happened? How does that compare to patients who made it all the way through the trial? Search for research about whether visit patterns can predict early safety signals.



**Databricks Genie (17/30) — Where it fell short**

The agent produced summary numbers (4.74 vs 5.52 visits) but the methodology is suspect: it used a natural-language Genie query rather than explicit SQL joining AE(seriousness='SERIOUS', discontinuation flag) → enrollment → visits counted before onset\_dt, meaning the critical data traps (using severity instead of seriousness, counting visits after onset, not anchoring to onset\_dt) may not have been avoided. The avg\_completed\_visits column shows 'nan' for distribution rows, suggesting the underlying query may be miscounted. The document search honestly reported no relevant findings but failed to leverage any hybrid structured+document synthesis.

**Q11** +9 **SF 27** vs **DBX 18** Split our trial portfolio in half chronologically — the older half versus the newer half by when trials started. Are we getting better or worse at enrolling patients over time? What do published studies show about global enrollment trends over the past decade?



**Databricks Genie (18/30) — Where it fell short**

The agent produced a plausible-looking analysis but delegated the chronological split and enrollment rate derivation entirely to Genie without verifying the SQL logic — the median split, screen failure denominator handling, and enrollment rate derivation are all opaque and potentially incorrect. The enrollment rate of ~80% appears suspiciously high and uniform, and the 'avg\_days\_first\_patient\_in\_to\_target\_enrollment\_achievement' column almost certainly does not exist natively, raising hallucination concerns. The document search was genuinely useful and grounded in corpus excerpts, but the structured data analysis lacks transparency on critical trap areas (E: enrollment rate derivation, J: screen failure denominator, T: temporal split methodology, Q: event vs patient grain).

**Q22** +9 **SF 26** vs **DBX 17** Which investigators have the best and worst track records in terms of bringing trials to successful completion versus having them terminated or withdrawn? Only consider investigators who have led at least 3 trials so we have a meaningful sample. Who are our top 5 and bottom 5? For the bottom 5, search PubMed for their publication history — are they publishing despite poor trial outcomes?



**Databricks Genie (17/30) — Where it fell short**

The agent retrieves top/bottom 5 investigators and presents them clearly, but there are significant data quality concerns. The bottom 5 all show 0% success rates with 0 completed trials, yet Dr. Sarah Thomas has 0 terminated/withdrawn trials out of 6 led — meaning all 6 are in some other status (likely ongoing/active), which would make a 0% 'success rate' misleading rather than indicative of poor performance. The agent fails to flag this critical distinction between 'not yet completed' and 'failed.' The publication search is inadequate — it relies only on the internal document corpus rather than actually querying PubMed, which the question explicitly requested.

**Q20** +8 **SF 26** vs **DBX 18** In how many of our trials does the experimental treatment arm have a higher serious AE rate than the placebo arm? Which drugs are the worst offenders? Search the drug labels for those drugs — do the labels warn about these safety signals?



**Databricks Genie (18/30) — Where it fell short**

The agent correctly identifies the key number (3,230 of 6,631 trials = 49%) and uses seriousness='SERIOUS' correctly (Trap A). However, the document search failed to find actual drug labels for the worst offenders — only generic class-level information was retrieved, and Lorlatinib label info came from clinical trial protocols, not actual drug labels. The agent did not clarify whether SAE rates are per-patient or per-event (Trap G), and the 'worst offenders' ranking by single-trial difference is statistically unreliable (small denominators, e.g., n=1 trial with extreme rates).

**Q21** +8 **SF 26** vs **DBX 18** Build a composite risk score for each trial by combining five factors: how far behind they are on enrollment targets, their visit compliance rate, whether their serious AE rate is above the portfolio average, how much they've exceeded their budget, and how many days their milestones are delayed. Which trials are the riskiest overall? For the top 3, search their published protocols for indicators of design complexity that might explain the problems.



**Databricks Genie (18/30) — Where it fell short**

The agent correctly used seriousness='SERIOUS' (not severity='SEVERE') and derived enrollment rate rather than using a non-existent column, which is good. However, the composite scoring methodology has significant flaws: the enrollment gap metric appears to be calculated as (target-actual)/target which would give ~96-98% for most trials rather than a meaningful differentiation, and many trials show 100% gap which is suspicious and likely a data artifact or calculation error. The document search for protocol complexity was largely unsuccessful - the knowledge base didn't contain the specific trial protocols, and the agent fell back on generic knowledge rather than grounded document findings, violating the hybrid requirement. The AE metric correctly uses seriousness but the scoring formula is non-standard and arbitrary.

**Q25** +8 **SF 26** vs **DBX 18** Rank our APAC countries by overall site performance, combining enrollment rates, visit compliance, and safety record (where fewer serious AEs is better) into a single normalized index. Which country has the best-performing sites? How does this compare to published benchmarks for site performance in Asia-Pacific clinical trials?



**Databricks Genie (18/30) — Where it fell short**

The agent correctly used seriousness='SERIOUS' (not severity='SEVERE') and applied MIN-MAX normalization with an inverted safety metric, which are positive signs. However, the enrollment rate calculation appears to use a simple enrolled/total ratio without properly accounting for screen failures in the denominator, and the visit compliance uses a fixed window rather than per-visit visit\_window\_days. The composite index for AU=0.00 is a mathematical artifact of min-max normalization that could mislead decision-makers into thinking AU has zero performance. The published benchmark comparison is weak and largely generic, not grounded in specific APAC clinical trial benchmarks from the document corpus.

**Q28** +7 **SF 26** vs **DBX 19** Prepare a red-flag report for our portfolio. I need three things: (1) which sites have both poor enrollment — below 60% of target — and excessive protocol deviations, more than 50 per 100 patients? (2) which drugs have a serious adverse event rate above 40% of patients AND have had regulatory submissions rejected? (3) which trials have both a budget overrun exceeding 20% and more than 3 milestones delayed? For each category of red flags, search published literature for remediation strategies.



**Databricks Genie (19/30) — Where it fell short**

The agent correctly identified the three red flag categories and ran structured queries, but committed critical data traps: (1) the enrollment rate query returned sites with rates ABOVE 60% (e.g., 61.5%, 63.7%), meaning the 983 'flagged' sites actually fail the <60% threshold — the query logic was inverted or the Genie tool misapplied the filter, producing a massive false positive list; (2) the SAE rate is computed as a percentage exceeding 100% (e.g., 622%), which indicates it is counting events per patient rather than patients with at least one SAE

— the agent failed to clarify or correct this AE grain trap, making the 458-drug count and all SAE rates unreliable; (3) the document search for budget/milestone remediation returned no relevant results and the agent acknowledged this, yet still presented generic advice as 'evidence-based.' The single trial finding (TRL-05658) appears plausible but cannot be trusted given the other methodological failures.

**Q13** +6 **SF 30** vs **DBX 24** How has our visit compliance rate changed quarter over quarter? Are patients showing up for their scheduled visits on time more often or less often as our portfolio matures? What published studies exist about the factors that drive visit adherence in long-running clinical trials?



 **Databricks Genie (24/30) — Where it fell short**

The agent correctly used visit\_window\_days in the compliance calculation (avoiding the fixed-window trap) and derived quarters from enrollment data. The quarterly compliance numbers are presented clearly and the document search was genuinely used. However, the ~49% compliance rate finding is suspicious and may indicate a data issue (e.g., random scheduling producing ~50% by chance), and the agent does not flag this as potentially anomalous or investigate further. The document search results appear somewhat generic and not deeply grounded in trial-specific corpus documents.

**Q07** +5 **SF 24** vs **DBX 19** Across our therapeutic areas, what is the ratio of medically serious adverse events to non-serious ones? Which therapeutic area has the highest proportion of serious events? For the worst area, look up the FDA drug labels for the drug classes we're using there — do the labels warn about high rates of serious events?



 **Databricks Genie (19/30) — Where it fell short**

The agent correctly identified Hematology as the highest serious AE area and used seriousness='SERIOUS' (not severity='SEVERE'), which is correct. However, the agent reports the ratio as a percentage (20.25%) rather than the expected ratio format (0.254), and the document search failed to find specific FDA label warnings for the actual drug classes used in Hematology trials — instead returning generic literature findings. The drug class breakdown for Hematology shows suspicious classes like 'SGLT2 Inhibitor' and 'GLP-1 Agonist' which are not typical hematology drugs, suggesting possible data quality issues the agent did not flag.

**Q24** +5 **SF 30** vs **DBX 25** How many of our trials have been running for more than two years past their planned end date and still haven't completed? How much of our total portfolio budget is tied up in these zombie trials? Search for published guidance on when sponsors should pull the plug on under-performing trials.



 **Databricks Genie (25/30) — Where it fell short**

The agent correctly identified 3,404 zombie trials and the actual spend of \$3.37B, matching the expected answer. However, it reported the planned budget as \$4.26B rather than correctly framing the \$3.37B as a percentage (33.9%) of the total portfolio budget (\$9.94B), which is the key metric requested. The document search was genuinely performed and grounded in corpus content about trial termination criteria. The methodology was sound — using status != COMPLETED and >730 days past end date — though the portfolio percentage calculation was missing.

**Q04** +4 **SF 30** vs **DBX 26** Across our five APAC markets, how do site quality scores compare? Which country has the most underperforming sites — specifically those rated below 3 out of 5? What does published research say about clinical trial site quality standards in Asia-Pacific?



 **Databricks Genie (26/30) — Where it fell short**

The agent correctly identified Japan as having the most below-3 sites (112) and produced country averages that closely match expected values (AU 3.81, JP 3.79, KR 3.75, SG 3.72, IN 3.67). The structured data query was well-executed and results are accurate. The document search returned limited relevant content but the agent was transparent about this limitation rather than hallucinating. Minor rounding differences in averages are negligible.

**Q26** +3 **SF 20** vs **DBX 17** Are there any drugs in our active trial portfolio that have never been mentioned in any PubMed abstract in our document collection? Are these truly novel compounds, or are they just under-studied? What is their regulatory submission status?



 **Databricks Genie (17/30) — Where it fell short**

The agent correctly identified that portfolio drugs have no PubMed presence and retrieved regulatory status data, but failed to perform a true SET DIFFERENCE operation — it sampled ~10 drugs rather than systematically identifying ALL drugs with no PubMed mentions. The document search methodology was weak: the agent never confirmed whether the document corpus actually contains PubMed abstracts indexed by these specific drug names, and the conclusion that these are 'synthetic test data' is speculative hallucination not grounded in any document evidence. The regulatory status table conflates submission-level rows with drug counts, creating confusion.

**Q02** +2 **SF 30** vs **DBX 28** How many patients experienced an adverse event that was medically serious but NOT clinically severe? What percentage of all patients is this? Search for FDA drug label examples where an AE can be serious from a regulatory standpoint but mild in terms of patient symptoms.



 **Databricks Genie (28/30) — Where it fell short**

The agent correctly identified 110,658 patients with SERIOUS but not SEVERE AEs, matching one of the expected key numbers exactly, and correctly used the seriousness column rather than severity. The percentage (37.6%) and total patient count (294,230) are provided with correct methodology. The document search acknowledged limitations honestly rather than hallucinating FDA label examples, though it did extract relevant protocol-based examples of the serious-but-mild concept.

**Q29** +2 **SF 26** vs **DBX 24** Here's a what-if scenario: if we took the wasted budget from our 10 most over-budget terminated trials and reinvested it into additional patient enrollment at our 10 best-performing sites, how many extra patients could we realistically enroll based on the per-patient cost at those sites? Would that meaningfully move the needle on our overall portfolio enrollment rate? Search for published case studies about budget reallocation and portfolio optimization in clinical development.



 **Databricks Genie (24/30) — Where it fell short**

The agent correctly executed all five sequential computations and produced plausible numbers with clear methodology. However, the enrollment rate calculation is suspect — using 400,023/500,000 looks like a round number artifact suggesting the denominator may not properly account for screen failures (Trap E/J), and the 'average cost per enrolled patient' derivation from site-level data may conflate trial-level costs with site-level costs without proper attribution. The document search was genuinely performed and grounded in corpus results, satisfying the hybrid requirement, though the case studies are loosely connected to the specific scenario.

**Q30** +2 **SF 26** vs **DBX 24** Prepare a board-level executive summary of our clinical portfolio with hard numbers, not just narratives. Cover: (1) portfolio health — enrollment rates by phase and country with trends over time, (2) safety — our serious AE rates versus what published drug labels cite as expected, (3) operations — which sites are performing best and worst based on quality scores and protocol adherence, (4) regulatory pipeline — what percentage of our submissions are getting approved and how long is the process taking, (5) research intelligence — the most important findings from published research relevant to our therapeutic areas. Give me specific numbers for everything.



 **Databricks Genie (24/30) — Where it fell short**

The agent produced a comprehensive, well-structured board summary with hard numbers from both structured data and document search, correctly using seriousness='SERIOUS' for AE classification. However, several data traps were triggered: the enrollment rate calculation appears to exclude screen failures from the denominator (using only screened patients rather than all screened including failures), the quarterly trend data showed suspicious 100%/0% enrollment rates that were correctly discarded in favor of a re-query, and the on-time visit rate of 49.3% was computed without confirming per-visit window logic. The document search for FDA label SAE comparisons returned no usable benchmarks, yet the agent still made comparative claims about portfolio rates 'aligning with expected ranges.'

**Q03** +1 **SF 15** vs **DBX 14** How many trials are currently in the recruiting phase versus those already actively dosing patients, broken down by therapeutic area? Which therapeutic areas have the biggest imbalance between recruitment and active treatment? What does published literature show about recruitment timelines in those areas?



 **Databricks Genie (14/30) — Where it fell short**

The agent's total ACTIVE count sums to 1957 and RECRUITING to 1490, which matches the expected numbers, but the per-area breakdown does not match the expected key numbers and the agent never explicitly reports the totals (1957 ACTIVE, 1490 RECRUITING) as required. The document search returned no specific results for the top imbalanced therapeutic areas, and the agent acknowledged this gap but still presented generic literature findings as if they were grounded answers. The methodology for identifying imbalance is reasonable but the absolute numbers per area cannot be verified against expected values.

**Q01** 0 **SF 30** vs **DBX 30** What percentage of patients are actually enrolled in our Phase 3 oncology trials running in Japan — based on where patients are physically located, not where the sponsor is headquartered? What do published studies say about oncology recruitment challenges specific to Japan?



 **Databricks Genie (30/30) — Where it fell short**

The agent correctly derived the 79.6% enrollment rate using the proper 3-table join with country\_cd='JP' (patient location, not sponsor), correctly computed enrolled/total patients, and avoided the screen failure trap by using the actual data grain. The document search surfaced Japan-specific oncology recruitment literature grounded in the corpus, covering regulatory, cultural, and infrastructure barriers.

|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |              |   |           |   |            |   |             |   |              |   |
|-----|----|-------|----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------|---|-----------|---|------------|---|-------------|---|--------------|---|
| Q06 | 0  | SF 30 | vs | DBX 30 | Beyond just being slightly outside the visit window, what percentage of patient visits are severely late — more than a week past the allowed window for that specific visit type? Which sites have the worst track record for severely late visits? What does published research say about how missed visit windows affect data quality? | Accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   | Groundedness |   | Relevance |   | Usefulness |   | Correctness |   | Consequences |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 | 5            | 5 | 5         | 5 | 5          | 5 | 5           | 5 | 5            | 5 |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 🔥 Databricks Genie (30/30) — Where it fell short                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |              |   |           |   |            |   |             |   |              |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | The agent correctly identified 59,034 severely late visits at 2.67% of completed visits (2,208,164), matching the expected key numbers precisely. The double derivation logic was correctly applied: first checking if a visit exceeds its allowed window, then checking if the excess exceeds 7 days. The site-level breakdown is well-structured, the document search was genuinely attempted with honest acknowledgment of gaps, and the overall methodology avoided common traps like using a fixed window instead of visit-specific windows.                                                                                                                     |   |              |   |           |   |            |   |             |   |              |   |
| Q14 | 0  | SF 21 | vs | DBX 21 | Are there any sites in our network that are actively running trials for drugs that have never been submitted to any regulatory authority? Does this represent a regulatory gap we should worry about? Search for what the regulatory requirements are in the countries where these sites operate.                                        | Accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   | Groundedness |   | Relevance |   | Usefulness |   | Correctness |   | Consequences |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 | 3            | 3 | 5         | 5 | 4          | 4 | 3           | 3 | 3            | 3 |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 🔥 Databricks Genie (21/30) — Where it fell short                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |              |   |           |   |            |   |             |   |              |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | The agent correctly identified the regulatory gap and found the right countries (AU, IN, JP, KR, SG), but reported 1,903 sites instead of the expected 1,967 (all statuses) or 1,390 (active/recruiting only). The agent's count of 1,903 appears to be a sum of the country-level distinct site counts (278+370+573+374+308=1,903), which may represent active-only sites but doesn't match either expected benchmark. The document search was attempted but yielded limited grounded regulatory content, with some generic statements about IND requirements. The Phase 4 without approval analysis is a reasonable inference but not fully grounded in the corpus. |   |              |   |           |   |            |   |             |   |              |   |
| Q23 | -1 | SF 23 | vs | DBX 24 | Among our patients on immunotherapy drugs who are also taking anticoagulant medications concurrently, is their serious AE rate higher than immunotherapy patients who are NOT on anticoagulants? What does the published literature say about known interactions between immunotherapy and anticoagulants?                               | Accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   | Groundedness |   | Relevance |   | Usefulness |   | Correctness |   | Consequences |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 | 4            | 4 | 4         | 5 | 5          | 4 | 3           | 4 | 3            | 3 |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 🔥 Databricks Genie (24/30) — Where it fell short                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |              |   |           |   |            |   |             |   |              |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | The agent correctly used seriousness='SERIOUS' (not severity='SEVERE'), which is the critical trap, and produced a clear comparison with specific numbers. However, the agent did not explicitly verify the conmed overlap logic (concurrent timing of anticoagulant use vs. immunotherapy), which is a key data trap — conmeds must overlap with the treatment period, not just exist in the record. The literature search was honest about its limitations but added limited value. The 5-table join complexity was handled implicitly but not transparently validated.                                                                                             |   |              |   |           |   |            |   |             |   |              |   |
| Q19 | -3 | SF 17 | vs | DBX 20 | Do we have any trials where not a single patient has any lab results on file? Is that because those trials don't require labs by design, or is it a data quality issue? Search for the study designs of these trials to determine if labs were expected.                                                                                 | Accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   | Groundedness |   | Relevance |   | Usefulness |   | Correctness |   | Consequences |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 | 3            | 3 | 3         | 4 | 3          | 4 | 3           | 3 | 2            | 3 |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | 🔥 Databricks Genie (20/30) — Where it fell short                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |              |   |           |   |            |   |             |   |              |   |
|     |    |       |    |        |                                                                                                                                                                                                                                                                                                                                          | The agent correctly identified that zero trials have no lab results, which matches the expected answer. However, it failed to compute the labs-per-patient ratio (the key metric requested), never identified TRL-05131 as having 7.3 labs/patient (the lowest), and instead focused on patient-level coverage percentage rather than the volume/density of lab results per patient. The document search was attempted but returned generic knowledge rather than trial-specific protocol information. The analysis is directionally useful but misses the core analytical requirement.                                                                               |   |              |   |           |   |            |   |             |   |              |   |